

MONTHLY FOOD GRAIN DISTRIBUTION ANALYSIS IN PUBLIC DISTRIBUTION SYSTEM (PDS)

1. Project Overview and Objective

Project Overview

This project focuses on analyzing food grain allocation and distribution data under India's Public Distribution System (PDS). The dataset contains monthly records of food grain allocation and distribution across states and districts.

Microsoft Excel was used for data cleaning, transformation, calculated field creation, and preliminary analysis. Power BI was used for data modeling, DAX calculations, and dashboard development to visualize distribution performance, allocation gaps, and grain-wise trends.

The project aims to transform raw government data into meaningful insights that support better planning and monitoring of food grain distribution.

Objectives

- To clean and transform raw food grain distribution data using Excel.
 - To analyze allocation and distribution patterns across states and districts.
 - To identify high-performing and low-performing regions.
 - To measure allocation gaps and distribution losses.
 - To evaluate monthly and quarterly distribution trends.
 - To create an interactive Power BI dashboard for decision-making.
 - To demonstrate Excel, Power BI, Data Modeling, and DAX skills.
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2. Data Source

Source Description

Public Distribution System (PDS) Food Grain Distribution Dataset

Source: https://indiadataportal.com/p/food-and-public-distribution/r/mocafpd-annavitrans_epos_transactions-dt-mn-aaa

Timeline

2017 – 2023

Domain

Government Welfare Analytics / Food Distribution Analytics

3. Problem Statement

The Public Distribution System (PDS) is responsible for ensuring food security by distributing essential food grains to beneficiaries throughout India.

The project addresses the following business problems:

- Analyze food grain distribution performance across states and districts.
 - Identify regions with the highest and lowest distribution efficiency.
 - Measure allocation gaps between allocated and distributed quantities.
 - Analyze food grain distribution trends over time.
 - Detect distribution losses and inefficiencies.
 - Support government authorities in improving distribution planning and monitoring.
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4. Attribute (Column) Details

Attribute Name	Data Type	Description
Id	Text	Unique Record Identifier
Date	Date	Distribution Date
state_name	Text	State Name
state_code	Numeric	State Code
district_name	Text	District Name
district_code	Numeric	District Code
total_qty_allocated	Decimal	Total Quantity Allocated
total_qty_distributed_epos	Decimal	Total Quantity Distributed Through ePOS
Distribution Percentage	Percentage	Distribution Efficiency
Allocation Gap	Decimal	Difference Between Allocation and Distribution
Distribution Loss	Percentage	Loss Percentage
Performance Category	Text	Excellent / Good / Moderate / Poor
Quarter	Text	Q1, Q2, Q3, Q4
Grains	Text	Rice, Wheat, Coarse Grain, Fortified Rice
Quantity	Decimal	Distributed Quantity
Type	Text	Allocation or Distribution

5. Tools and Technologies

Microsoft Excel

Used for:

- Data Cleaning
- Handling Missing Values
- Removing Duplicate Records
- Data Transformation
- Calculated Fields
- Filtering and Sorting
- Pivot Tables
- Initial Data Analysis

Power BI

Used for:

- Data Modeling
 - Relationship Creation
 - DAX Calculations
 - Dashboard Design
 - KPI Development
 - Interactive Reporting
 - Trend Analysis
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6. Data Pre-Processing (Excel / Power Query)

Data Cleaning

The following cleaning activities were performed:

- Removed duplicate records.
- Checked and handled missing values.
- Standardized date formats.
- Verified state and district names.
- Removed inconsistent records.
- Validated numeric fields.

Data Transformation

The following calculated columns were created:

Distribution Percentage

Distribution Percentage=IFERROR([@[total_qty_distributed_epos]]/[@[total_qty_allocated]],0)

Allocation Gap

Allocation Gap = [@[total_qty_allocated]]-[@[total_qty_distributed_epos]]

Distribution Loss

Distribution Loss = IFERROR([@[Allocation Gap]]/[@[total_qty_allocated]],0)

Quarter

Quarter ="Q"&ROUNDUP(MONTH([@month])/3,0)

Performance Category

Performance Category = IF([@[Distribution Percentage]]>=95%,"Excellent",IF([@[Distribution Percentage]]>=80%,"Good",IF([@[Distribution Percentage]]>=65%,"Moderate","Poor")))

Filtering and Sorting

- Sorted by State, District and Date.
- Filtered records for performance analysis.

Pivot Tables Created

1. State-wise Distribution Summary
2. Monthly Distribution Trend
3. Performance Category Analysis

Unpivot Columns

- The grain distribution columns (**Rice, Wheat, Coarse Grain, and Fortified Rice**) were unpivoted in Power Query.
 - This transformation converted multiple grain-specific columns into two attributes:
 - **Grains** (Rice, Wheat, Coarse Grain, Fortified Rice)
 - **Type** (Allocated,Distributed)
 - **Quantity** (Corresponding quantity values)
 - The unpivot operation transformed the data from a **wide format** to a **long format**, making it more suitable for Power BI reporting, filtering, and grain-wise analysis.
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7. Data Modeling and DAX (Power BI)

Data Model

A Star Schema model was implemented.

Fact Table

Fact Food Distribution

Contains:

- Allocation Data
- Distribution Data
- Grain Details
- Performance Metrics

Dimension Tables

Dim State

- State Name
- Total States

Dim District

- district_code
- district_name
- State Name

Dim Date

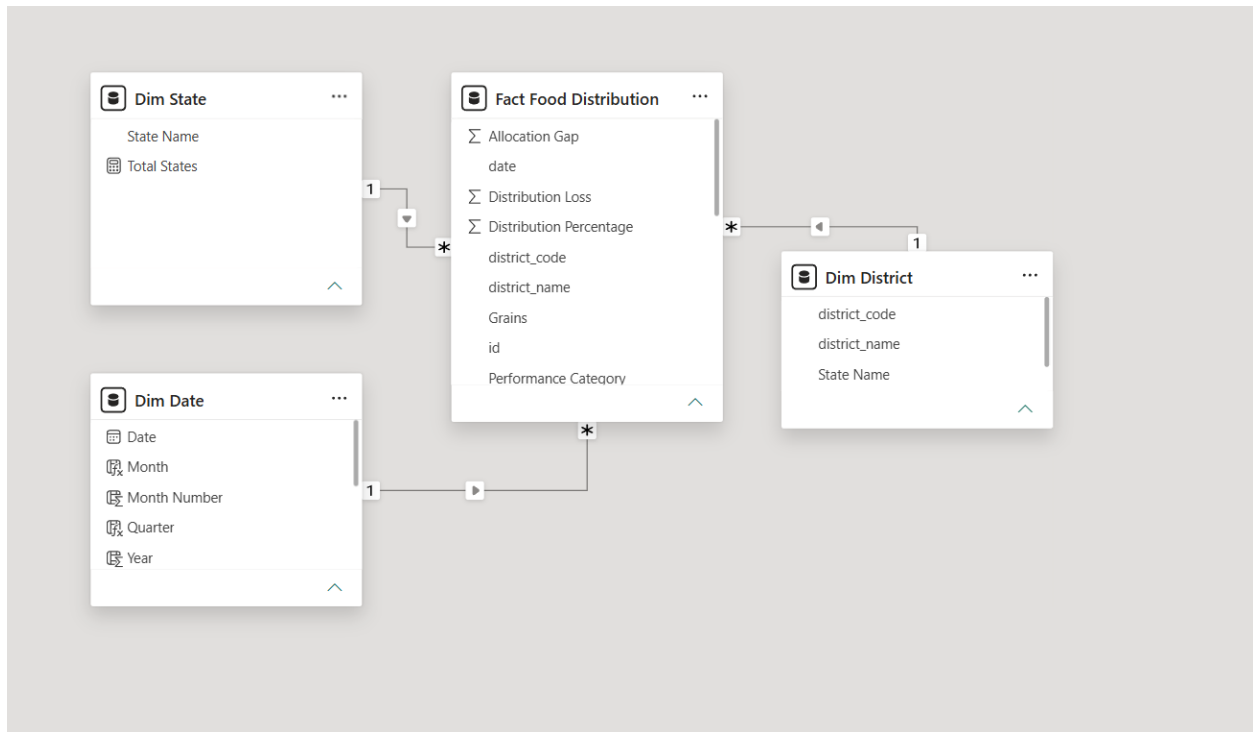
- Date
- Month
- Month Number
- Quarter
- Year

Relationships

Dimension Table	Fact Table	Relationship
Dim State	Fact Food Distribution	State Name
Dim District	Fact Food Distribution	District Code
Dim Date	Fact Food Distribution	Date

Cardinality

- One-to-Many Relationships
- Single Direction Filtering



DAX Measures

Total Allocation

Total Allocation = `CALCULATE(SUM('Fact Food Distribution'[Quantity]),'Fact Food Distribution'[Type] = "Allocated")`

Total Distribution

Total Distribution = `calculate(sum('Fact Food Distribution'[Quantity]),'Fact Food Distribution'[Type]="Distributed")`

Total Allocation Gap

Allocation Gap = `[Total Allocation]-[Total Distribution]`

Average Allocation

Average Allocation = `AVERAGE('Fact Food Distribution'[Quantity])`

Distribution Efficiency %

Distribution % = `DIVIDE([Total Distribution],[Total Allocation],0)`

Total States

Total States = `Count('Dim State'[State Name])`

Total Districts

`DISTINCTCOUNT(district_name)`

MTD Distribution

MTD Distribution = TOTALMTD([Total Distribution],'Dim Date'[Date])

YTD Distribution

YTD Distribution = TOTALYTD([Total Distribution],'Dim Date'[Date])

Top Performing State

Top Performing State = TOPN(1,VALUES('Fact Food Distribution'[state_name]),[Total Distribution])

8. Analysis and Visualizations (Power BI)

Dashboard Overview

An interactive dashboard was developed to provide comprehensive insights into food grain allocation and distribution performance.

KPI Cards

- Total Food Grain Allocated
- Total Food Grain Distributed
- Distribution Efficiency %
- Allocation Gap
- Distribution Loss %
- Total States Covered

Visualizations Used

Top 10 States by Distribution

Clustered Bar Chart showing distributed quantities by top 10 states.

Monthly Distribution Trend

Line Chart showing month-wise distribution trends.

Top 10 States by Distribution

Stacked Bar Chart showing distributed quantities by top 10 states.

Distribution by Grains and Type

Clustered Column Chart displaying allocated and distributed quantities of food grains.

Distribution by Performance Category

Pie Chart displaying Excellent, Good, Moderate, and Poor categories.

Year-wise Distribution Analysis

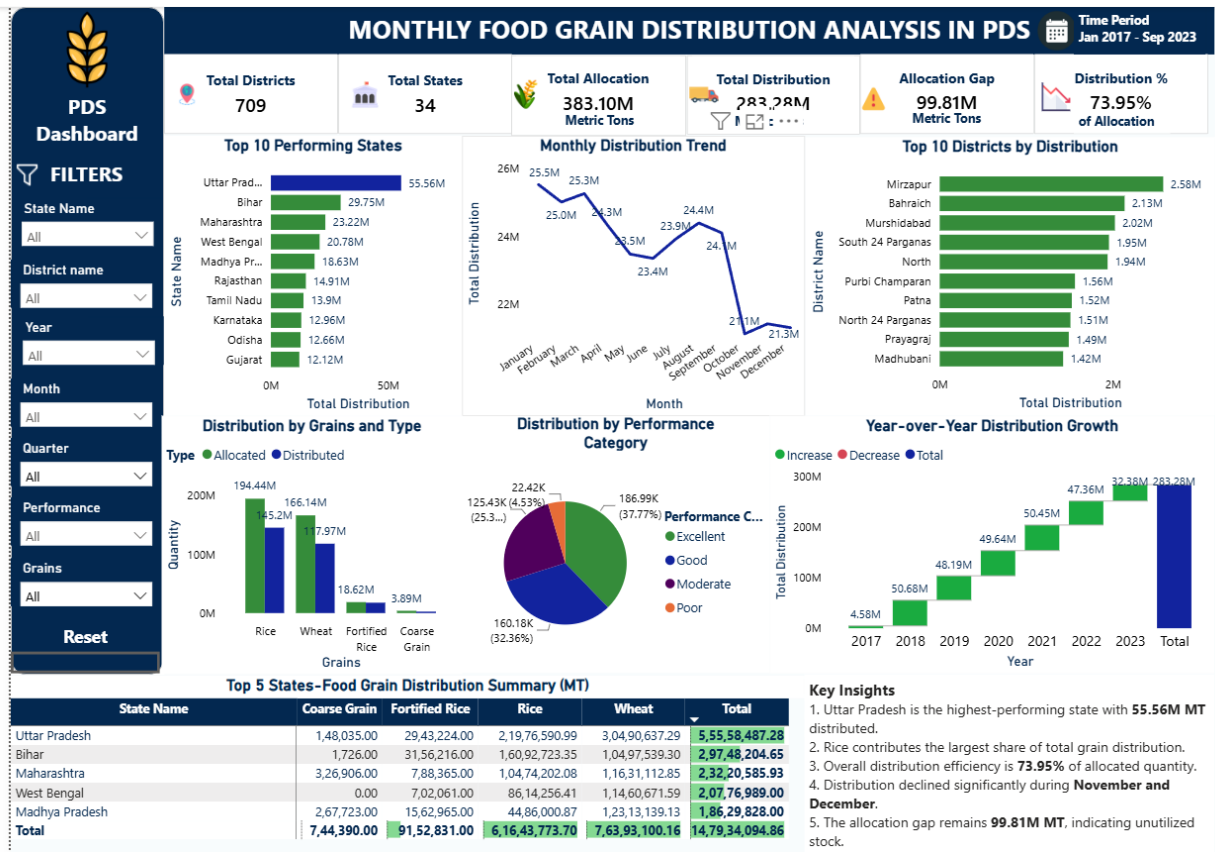
Waterfall chart is displaying how each year's distribution contributes to the overall total.

State-wise Food Grain Matrix summary

Matrix/Table showing state-wise top 5 grains performance.

Interactive Features

- State Slicer
- District Slicer
- Year Slicer
- Month Slicer
- Quarter Slicer
- Performance Slicer
- Grains Slicer
- Drill Down
- Cross Filtering
- Dynamic Titles
- Icons
- Buttons
- Bookmarks



- Monitor allocation gaps regularly.
- Improve logistics in districts showing recurring losses.
- Use dashboard KPIs for proactive monitoring.
- Allocate resources based on performance trends.

Business Recommendations

- Focus on reducing the allocation gap of 99.81M MT.
 - Investigate the decline in distribution during November and December.
 - Replicate best practices from high-performing states such as Uttar Pradesh.
 - Increase monitoring of districts with lower distribution efficiency.
 - Expand ePOS adoption to improve transparency and reduce losses.
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10. Conclusion

The Monthly Food Grain Distribution Analysis project successfully demonstrated the application of Excel and Power BI for end-to-end data analytics.

Excel was utilized for data cleaning, transformation, creation of calculated fields, and preliminary analysis. Power BI was then used for advanced data modeling, DAX calculations, KPI development, and the creation of an interactive dashboard.

The resulting dashboard provides meaningful insights into food grain allocation, distribution efficiency, allocation gaps, grain-wise performance, and regional trends. These insights can support government agencies in improving distribution planning, reducing wastage, and strengthening the effectiveness of the Public Distribution System (PDS).

Overall, this project highlights how data analytics and business intelligence tools can convert raw government data into actionable insights that enable better decision-making and operational efficiency
